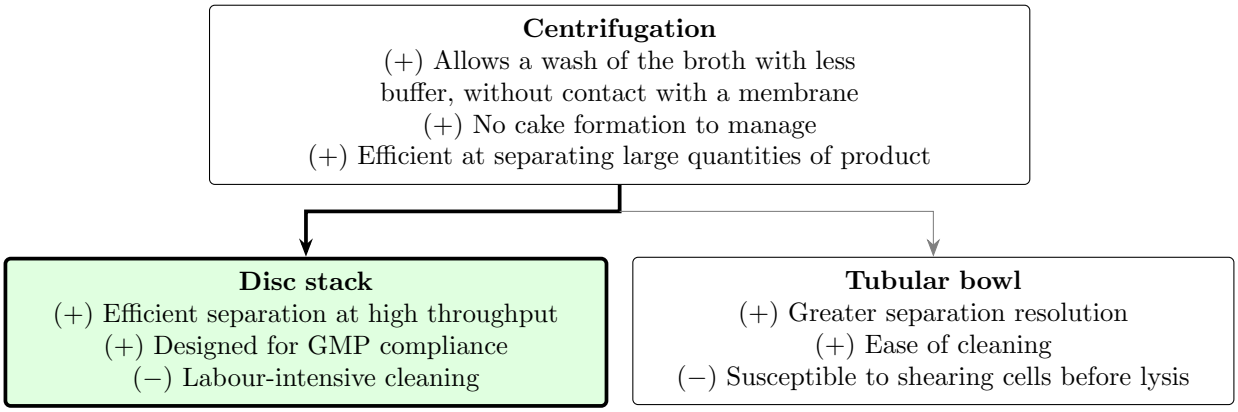
Centrifuge Wash Cycle: Stream Balances and Processing Time

UBC CHBE capstone · Montbretin A production · Biomass separation (PFD 400): T-401, P-402, C-401

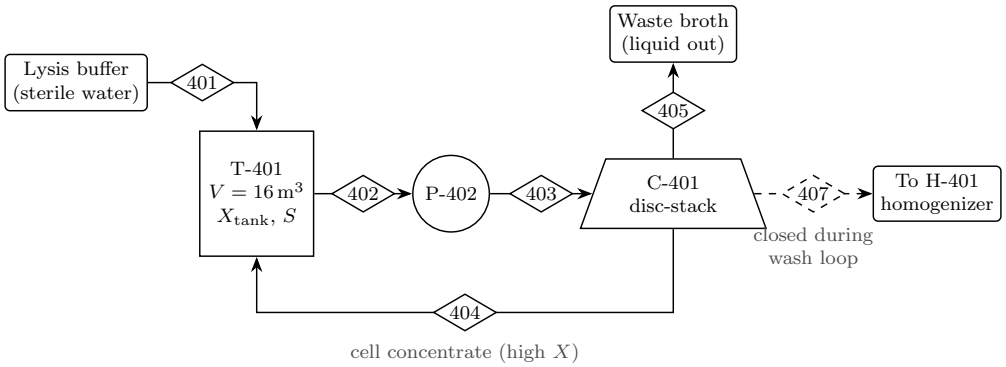
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Equipment selection



Selected option shaded. A disc-stack centrifuge (C-401) handles the full 16 m^3 harvest at high throughput; the cleaning burden is the trade-off. Adapted from the capstone report, Figure A2.

Process diagram



Idea: replace the spent broth liquid with lysis buffer (sterile water, stream 401) before lysis. The centrifuge sends the cells back to the tank in the concentrate (404) and the liquid out as waste (405); fresh buffer replaces the waste one-for-one, so the tank volume stays constant.

Symbols: F_i volumetric flow of stream i [m^3/h]; X_i cell concentration [g DCW/L]; S fraction of tank liquid that is fresh buffer; V tank liquid volume; $D_i = F_i/V$ dilution rate [1/h]; n number of passes (tank turnovers).

Given

Tank volume, V	$16\,000\,\mathrm{L} = 16\,\mathrm{m}^3$ (harvested broth)
Cells, $X_{\mathrm{tank}} = X_{402} = X_{403}$	$22.647\,\mathrm{g/L}$ DCW (end of fermentation)
Feed to centrifuge, F_{403}	$16\,\mathrm{m}^3$ in $2\,\mathrm{h} \Rightarrow 8\,\mathrm{m}^3/\mathrm{h}$ (first basis)
Cell recovery per pass	$F_{404}X_{404} = 0.999\,F_{403}X_{403}$ (99.9 % to concentrate)
Liquid split	$F_{405} = 0.7\,F_{403}$ (70 % of liquid to waste)
Buffer, stream 401	$X_{401} = 0\,\mathrm{g/L}, \quad S_{401} = 1$ (pure buffer)
Wash target	replace 99.9 % of the broth liquid with buffer: $S = 0.999$
Centrifuge (C-401)	Alfa Laval disc-stack, $35\,\mathrm{m}^3/\mathrm{h}$ max, $55\,\mathrm{kW}$

Find

(1) F_{401}, F_{404}, F_{405} and X_{404}, X_{405} to fully define the centrifuge loop; (2) the processing time and number of passes to replace 99.9 % of the broth liquid; (3) the same at the centrifuge's maximum flow, $35\,\mathrm{m}^3/\mathrm{h}$.

Equations

Steady flow around the centrifuge (in − out + gen − consumed = accumulation, with no generation, consumption or accumulation in C-401):

$$F_{403} = F_{404} + F_{405} \tag{1}$$

$$F_{403}X_{403} = F_{404}X_{404} + F_{405}X_{405} \tag{2}$$

Buffer fraction in the well-mixed tank (streams 404 and 403 carry tank liquid, so $S_{404} = S_{403} = S$):

$$V\frac{dS}{dt} = F_{404}S_{404} + F_{401}S_{401} - F_{403}S_{403} \tag{3}$$

$$\frac{dS}{dt} = 0.7\,D_{403}\,(S_{401} - S), \qquad D_{403} = \frac{F_{403}}{V} \tag{4}$$

$$t = \frac{-\ln\left(\dfrac{S_{401} - S}{S_{401} - S_0}\right)}{0.7\,D_{403}}, \qquad n = \frac{F_{403}\,t}{V} \tag{5}$$

Assumptions

- Constant tank volume: buffer in equals waste out, $F_{401} = F_{405}$. The waste flow from one pass is the buffer flow into the next.
- Tank well mixed; the centrifuge separates cells, not dissolved species, so the concentrate (404) and waste (405) liquids have the tank's buffer fraction S.
- No buffer in the tank at the start: $S_0 = 0$.
- Stream 407 (to the homogenizer) is closed during the wash loop.
- Cell concentrations below use the first-pass value $X_{403} = 22.647\,\mathrm{g/L}$. The tank loses only about 1 % of its cells over the whole wash (see Check), so this is the largest X_{405}.
- Two different 99.9 % figures: the cell recovery per pass, and the wash target for the liquid.

Worked example

1. Liquid flows (basis $F_{403} = 8\,\mathrm{m^3/h}$)

$$F_{405} = 0.7\,F_{403} = 0.7\,(8\,\mathrm{m^3/h}) = \boxed{5.6\,\mathrm{m^3/h} = F_{401}}$$

$$F_{404} = F_{403} - F_{405} = (1-0.7)\,F_{403} = 0.3\,(8\,\mathrm{m^3/h}) = \boxed{2.4\,\mathrm{m^3/h}}$$

2. Cell concentrations

Waste, substituting $F_{404}X_{404} = 0.999\,F_{403}X_{403}$ and $F_{405} = 0.7\,F_{403}$ into equation (2):

$$F_{403}X_{403} - 0.999\,F_{403}X_{403} = 0.7\,F_{403}X_{405}$$

$$\Rightarrow\; X_{405} = \frac{(1-0.999)\,X_{403}}{0.7} = \frac{(22.647\,\mathrm{g/L})(0.001)}{0.7} = \boxed{0.03235\,\mathrm{g/L\ DCW}}$$

Concentrate, with $F_{403} = F_{404}/0.3$:

$$X_{404} = \frac{0.999\,F_{403}X_{403}}{F_{404}} = \frac{0.999\,X_{403}}{0.3} = \frac{(0.999)(22.647\,\mathrm{g/L})}{0.3} = \boxed{75.41\,\mathrm{g/L\ DCW}}$$

Stream	401	402	403	404	405
F [$\mathrm{m^3/h}$]	5.6	8	8	2.4	5.6
X [$\mathrm{g\,DCW/L}$]	0	22.647	22.647	75.41	0.03235

3. Processing time: continuous recycle (Method 1)

Substituting $F_{404} = 0.3\,F_{403}$, $F_{401} = 0.7\,F_{403}$ and $S_{404} = S_{403} = S$ into equation (3):

$$\frac{dS}{dt} = 0.3\,D_{403}S + 0.7\,D_{403}S_{401} - D_{403}S = D_{403}\,[0.3S - S + 0.7S_{401}] = 0.7\,D_{403}\,(S_{401} - S)$$

$$\int_0^{0.999} \frac{dS}{S_{401} - S} = \int_0^t 0.7\,D_{403}\,dt \;\Rightarrow\; -\ln\!\left(\frac{1-0.999}{1-0}\right) = 0.7\,D_{403}\,t$$

With $D_{403} = \dfrac{F_{403}}{V} = \dfrac{8\,\mathrm{m^3/h}}{16\,\mathrm{m^3}} = 0.5/\mathrm{h}$ and $S_{401} = 1$:

$$t = \frac{-\ln(0.001)}{0.7\,(0.5/\mathrm{h})} = \frac{6.908}{0.35/\mathrm{h}} = \boxed{19.74\,\mathrm{h}}$$

Too long for one batch, so the feed rate was iterated up to the centrifuge's limit.

4. Iterating the feed rate

t_{pass} is the time to pass the $16\,\mathrm{m^3}$ once through the centrifuge, so $F_{403} = V/t_{\mathrm{pass}}$; then $F_{401} = F_{405} = 0.7\,F_{403}$ and $F_{404} = 0.3\,F_{403}$.

t_{pass} [h]	F_{403} [m^3/h]	D_{403} [1/h]	t [h]
2.0	8.00	0.500	19.74
1.9	8.42	0.526	18.75
1.8	8.89	0.556	17.76
1.7	9.41	0.588	16.78
1.6	10.00	0.625	15.79
1.5	10.67	0.667	14.80
1.4	11.43	0.714	13.82
1.3	12.31	0.769	12.83
1.2	13.33	0.833	11.84
1.1	14.55	0.909	10.86
1.0	16.00	1.000	9.87
0.8	20.00	1.250	7.89
0.7	22.86	1.429	6.91
0.64	25.00	1.563	6.32
0.6	26.67	1.667	5.92
0.5	32.00	2.000	4.93
0.457	**35.00**	**2.188**	**4.51**

5. Design point: $35\,\text{m}^3/\text{h}$ (C-401 maximum)

$$D_{403} = \frac{35\,\text{m}^3/\text{h}}{16\,\text{m}^3} = 2.1875/\text{h}, \qquad t = \frac{-\ln(0.001)}{0.7\,(2.1875/\text{h})} = \boxed{4.511\,\text{h}}$$

$$n = \frac{F_{403}\,t}{V} = \frac{(35\,\text{m}^3/\text{h})(4.511\,\text{h})}{16\,\text{m}^3} = 9.868 \;\Rightarrow\; \boxed{10\text{ passes}}, \qquad \text{cells retained} = 0.999^{10} = \boxed{99.00\,\%}$$

Design flows: $F_{401} = F_{405} = 24.5\,\text{m}^3/\text{h}$, $F_{404} = 10.5\,\text{m}^3/\text{h}$; as mass, streams 401, 403, 404 and 405 carry 24 426, 35 070, 10 521 and 24 549 kg/h (water 997, broth $1002\,\text{kg}/\text{m}^3$, so 401 and 405 match in volume, not mass). X_{404} and X_{405} are unchanged.

Check

- **Passes are independent of flow:** $n = D_{403}t = -\ln(0.001)/0.7 = 9.868$ in every row of the table.
- **Buffer per batch:** $(24.5\,\text{m}^3/\text{h})(4.511\,\text{h}) = 110.5\,\text{m}^3 = -\ln(0.001)\,V$, i.e. 6.908 tank volumes at any flow.
- **Cell loss, continuous form:** $V\,dX/dt = -0.001\,F_{403}X \Rightarrow X/X_0 = e^{-0.001\,n} = 99.02\,\%$, matching $0.999^{10} = 99.00\,\%$ to within rounding the passes up.